Change force while total mass stays fixed

LEARNING TARGET · S4 I can test $F = ma$ while keeping total mass fixed.

A · KEY VOCABULARY

Driving force	The pull before resistance is subtracted.
Net force	Driving force minus opposing resistance.
Slope	On an a-versus-net-force graph: $1/M$.
Offset	Fixed friction shifts the driving-force line down.

B · KEY EQUATIONS + UNITS

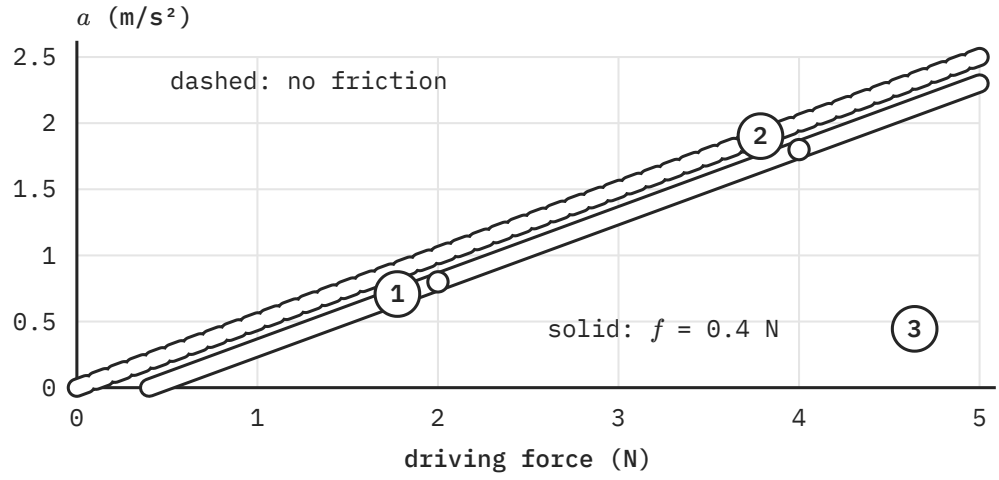
$$a = F_{\text{net}}/M$$

$$F_{\text{net}} = F_{\text{drive}} - f$$

M total mass, kg
N/kg equals m/s²

C · COLOR IT · DRIVING VS. NET FORCE

*Moving +; M = **2 kg**.*



☐ **1 · PENCIL** Axes and data

☐ **2 · GREEN** Speeding up here

☐ **3 · ORANGE** Force labels

D · WORKED EXAMPLE

A **2 kg** system slides with **0.4 N** of opposing friction. **Compare acceleration with driving forces of 2 N and 4 N.**

GIVEN	EQUATION	WORK	ANSWER
M = **2 kg** f = **0.4 N** F_{drive} = **2 N**, then **4 N**	Subtract friction. Then $a = F_{\text{net}}/M$.	$a_1 = (2 - 0.4)/2 = 0.8$ $a_2 = (4 - 0.4)/2 = 1.8$	**0.8; 1.8 m/s²** Driving force doubles; a does not.

E · THE 2026-XJ THREAD

DAY 12 OF 22 · NEW TOOL: NET-FORCE TEST

A proposed push on XJ must be added to every other external force. Only the net force belongs in the calculation of acceleration.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else